



< Previous





Next >

9.2.5 Diagonalizing a matrix

 Bookmark this page

Pursue a verified certificate



Verified Certificate

 THE UNIVERSITY of TEXAS SYSTEM

This is to certify that

Sandipan Dey

successfully completed and received a passing grade in

UT.ALA:Advanced Linear Algebra: Foundations to Frontiers

a course of study offered by UTAustinX, an online learning initiative of University of Texas System.




Maurilio Pugliesi
Professor
UniversityX


Justine Doe, Ph.D.
Professor
UniversityX



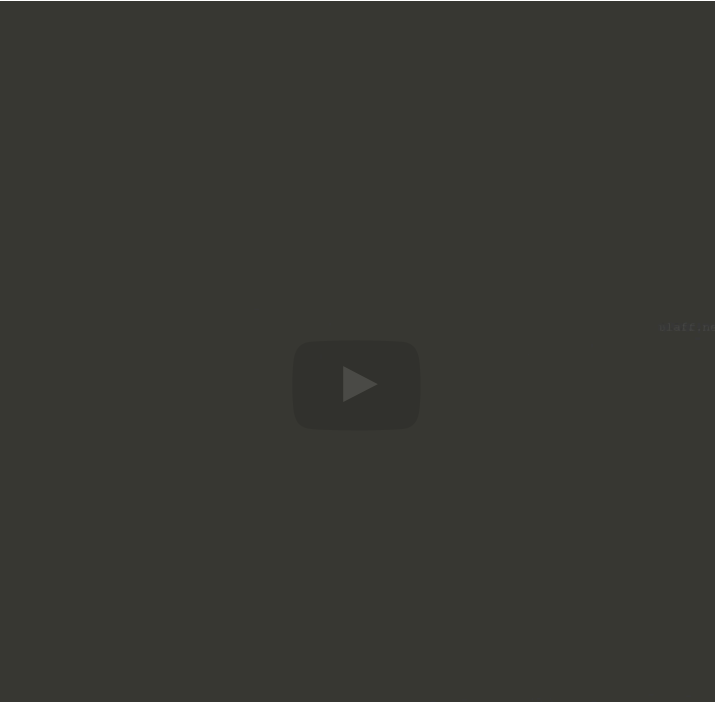
Verified Certificate
Issued Month / Year

Valid Certificate ID
12345678901234567890

- Unlimited access to all course materials
- Shareable [verified certificate](#)
- AI learning assistant and translations

Upgrade for \$99

Video



▶ 3:27 / 3:27

▶ 2.0x 🔊 📺 📄

x_1 , and so forth, so here we get λ_{n-1} times x_{n-1} . And on the left we know that multiplying a matrix times the individual columns just means multiplying that matrix times individual columns. And if we now set columns on the left equal to columns on the right, what we get is exactly the A times x_0 is equal to λ_0 times x_0 and so forth. So from this, we conclude that if we have such a matrix X then

Video

📄 [Download video file](#)

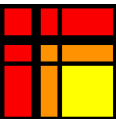
Transcripts

📄 [Download SubRip \(.srt\) file](#)

📄 [Download Text \(.txt\) file](#)

Reading assignment

1/1 point (graded)



Read Unit 9.2.5 of the notes. [\[LINK\]](#)

☒ done



Submit

✔ Correct (1/1 point)

Discussion

Hide Discussion

Topic: Week 09 / 09.2.5

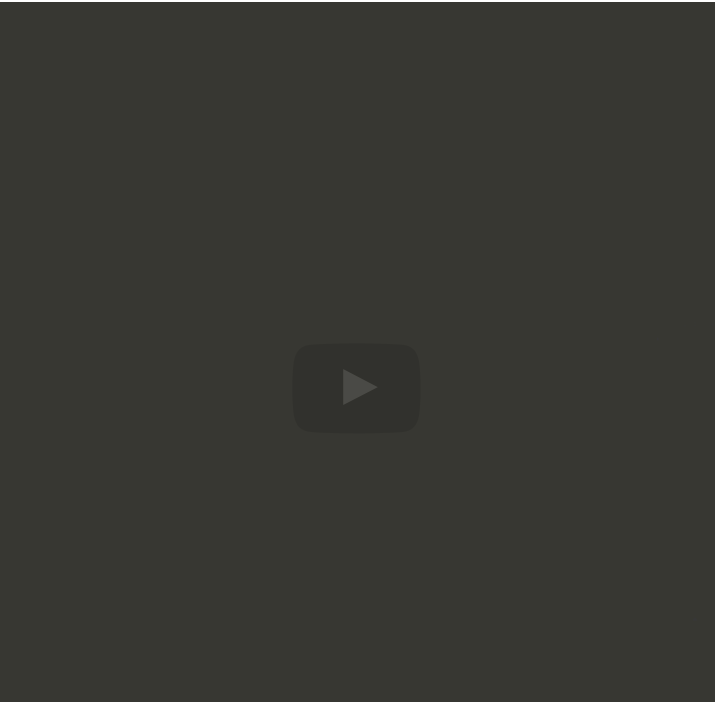
Add a Post

Show all posts ▼ by recent activity ▼

There are no posts in this topic yet.

✕

Video



▶ 5:01 / 5:01

▶ 2.0x 🔊 🔍 CC “

vector $\tilde{v}$. So what does that really mean? All we've done here is we've replaced v with X times $\tilde{v}$, where we now in $\tilde{v}$ have the coefficients in terms of this basis that we get from the columns of our matrix X . And we can obviously do the exact same thing to w . We can say, "Let's also view that vector in the same basis." And then this here becomes the vector $\tilde{w}$. So what do we have now? We have

Video

📄 [Download video file](#)

Transcripts

- 📄 [Download SubRip \(.srt\) file](#)
- 📄 [Download Text \(.txt\) file](#)

Homework 9.2.5.1

1/1 point (graded)
In Homework 9.2.3.3, we computed the eigenvalues and corresponding eigenvectors of

$$A = \begin{pmatrix} -2 & 3 & -7 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix}.$$

Use the answer to that question to give a matrix X such that $X^{-1}AX = \Lambda$. Check that $AX = X\Lambda$.

☒ done



Solution:

For a complete solution see the [notes](#).

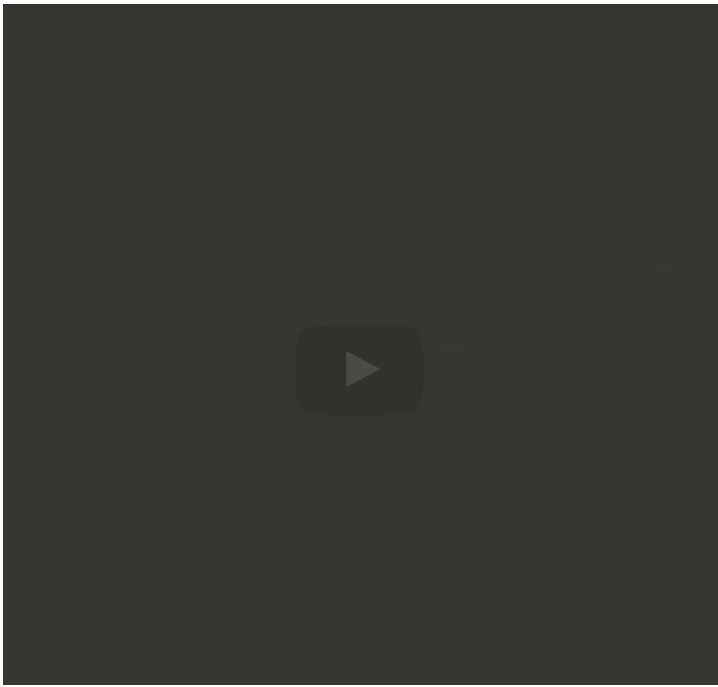
Submit

🔔 Answers are displayed within the problem

Video



raising this to the k th power. And if you think about it, if you multiply these



▶

2:10 / 2:10

▶ 2.0x

🔊

🔍

CC

“

Video

📄 [Download video file](#)

Transcripts

📄 [Download SubRip \(.srt\) file](#)

📄 [Download Text \(.txt\) file](#)

Homework 9.2.5.2

1/1 point (graded)
Let $A \in \mathbb{C}^{m \times m}$ have distinct eigenvalues.

ALWAYS/SOMETIMES/NEVER: A is diagonalizable.

ALWAYS

▼

✔ Answer: ALWAYS

< Previous

Next >

multiply these together the X inverse cancels the X every time. And therefore this is actually the same as A times Λ to the k th power times X inverse times v . Now raising a diagonal matrix to the k th power is easy. That's a matter of raising each of the individual diagonal elements to the k th power. So now what you can do is if you want to apply A to v k times, you can



edX

- [About](#)
- [Affiliates](#)
- [edX for Business](#)
- [Open edX](#)
- [Careers](#)
- [News](#)

Legal

- [Terms of Service & Honor Code](#)
- [Privacy Policy](#)
- [Accessibility Policy](#)
- [Trademark Policy](#)

Connect

- [Idea Hub](#)
[Contact Us](#)
[Help Center](#)
[Security](#)
[Media Kit](#)

